Diagnosing Receiver-Sensitive Out-of-Band Dependencies in LoRa Radio-Frequency Fingerprint Identification

Ziyang Wang, Chengcheng Han, Jianhuan Chen, and Xin Zhang, *Member, IEEE*

***Abstract*—Out-of-band (OOB) distortion can aid radio-frequency fingerprint identification (RFFI), but its discriminative value does not guarantee stability across receivers. We investigate receiver-sensitive OOB dependence in closed-set, cross-receiver Long Range (LoRa) RFFI using interventions on frozen models, spectral-pathway controls, and separate receiver sets for development and blind evaluation. Two diagnostic models incorporate OOB information differently: a four-view convolutional late-fusion model and a token-based model with OOB cross-attention. We modify targeted OOB inputs through amplitude scaling, shuffling across devices recorded by the same receiver, neutral replacement, and attenuation of either spectral side, while keeping model parameters and other inputs fixed. Both models depend on the targeted OOB pathway across all six blind receivers. After averaging the shuffle and neutral-replacement drops within each receiver, the median drops across the six blind receivers are 49.05 percentage points for LateFusion-OOB and 71.78 percentage points for CrossAttn-OOB. The corresponding median drops under the most disruptive tested OOB amplitude scale are 51.83 and 70.66 percentage points. Left-side attenuation causes larger drops than right-side attenuation on every blind receiver. Controls based on channel-independent spectrograms and receiver-adversarial training achieve accuracies of 96.41–100.00% on the blind receivers. On the development folds, random OOB-scale augmentation reduces the mean accuracy drop caused by halving the OOB amplitude from 53.96 to 2.92 percentage points but does not improve held-out accuracy relative to matched continued training without augmentation. These results show that useful OOB features can remain receiver-sensitive and that robustness to a specified OOB perturbation does not establish receiver invariance.**

***Index Terms*—cross-receiver generalization, intervention analysis, LoRa, out-of-band spectrum, radio-frequency fingerprint identification, receiver sensitivity.**

I. Introduction

RADIO-FREQUENCY (RF) fingerprint identification (RFFI) distinguishes transmitters through repeatable imperfections introduced by oscillators, mixers, filters, and power amplifiers. Because identification is performed at the receiver and requires no endpoint modification, RFFI is attractive for resource-constrained Internet of Things (IoT) deployments [1]–[4]. Long Range (LoRa) presents a challenging setting: signals from low-power devices may be received by heterogeneous gateways, and transmitter-dependent spectral features may extend beyond the nominal 125-kHz band.

RFFI systems have used raw in-phase/quadrature (I/Q) samples, spectrograms, and hardware-related spectral features. WideScan showed that combining in-band (IB) and out-of-band (OOB) distortion can improve discrimination among similar transmitters [5].

However, OOB information that is useful under one acquisition setting may not remain stable when the receiver changes. A captured fingerprint reflects transmitter hardware, propagation, receiver front-end response, sampling, and collection conditions. Classifiers may exploit features associated with time and channel conditions as shortcuts [6]–[8], while receiver gain, filtering, clocking, and digitization can introduce further distribution shifts. These effects are present before a digital IB or OOB view is constructed. An OOB feature may therefore aid identification while varying systematically across receivers.

Cross-receiver RFFI studies have addressed this problem using receiver-adversarial learning, domain alignment or adaptation, multi-receiver training, and contrastive representations [9]–[16]. Studies of OOB features have focused on the benefits of additional spectral information for transmitter discrimination. The studies reviewed above do not directly establish which OOB features a trained classifier uses, whether its dependence on those features is receiver-sensitive, or whether that dependence persists on receivers excluded from model and protocol development.

We address this gap by diagnosing how trained classifiers depend on OOB information across receivers. We analyze transmitter- and receiver-associated OOB features and intervene on frozen classifiers. We compare the intervention effects with spectral-pathway controls and models that assess task solvability, then test predefined directional criteria on six blind receivers. The diagnostic models are a four-view convolutional late-fusion model and a token-based model with OOB cross-attention.

The main contributions are as follows.

1) We formulate receiver-sensitive OOB dependence and distinguish OOB utility, frozen-model sensitivity, and transfer to unseen receivers. Intervention targets are defined separately for the two representation families.
2) We develop four OOB interventions for frozen classifiers: amplitude scaling, same-receiver cross-device shuffling,

Ziyang Wang and Chengcheng Han are with the School of Electronic and Information Engineering, Beihang University, Beijing, China (e-mail: wangziyang1@126.com, HAN106124@163.com).

J. Chen is with Guangxi Power Grid Co., Ltd., Nanning 530023, Guangxi, China (e-mail: c_jh.xt@gx.csg.cn).

Xin Zhang is with the School of Cyber Science and Technology, Beihang University, Beijing, China (e-mail: zhangxinn@buaa.edu.cn).

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neutral replacement, and spectral-side attenuation. A raw-spectrum in-band control (TrueIB) assesses changes to the full feature-generation pathway, while task-solvability controls assess whether identification remains feasible on held-out receivers.

3) Across two diagnostic models, we find that OOB features support transmitter discrimination while retaining receiver-associated information. The models' dependence on these features varies across receivers and spectral regions, and the benefit of OOB information differs between representations.
4) We fix the protocol before blind evaluation. The predefined directional effects occur on all six blind receivers. In the development experiments, reducing OOB-scale sensitivity does not improve accuracy on held-out receivers relative to matched clean continuation.

The results do not support a blanket recommendation to retain or remove OOB evidence. OOB discriminability and receiver invariance are distinct properties of a learned representation, and robustness to one engineered OOB perturbation cannot substitute for evaluation on unseen receivers.

Section II reviews prior work. Sections III–VI present the formulation, method, protocol, and results. Section VII discusses the findings, and Section VIII concludes the paper.

II. Related Work

A. RF Fingerprints and Deployment Variation

Radiometric methods and deep classifiers have established that analog transmitter imperfections can support device identification [1]–[3], [17]. LoRa studies have used raw I/Q samples, spectrograms, coherent integration, channel-independent representations, and multi-condition datasets [18]–[24]. Measurements across time, location, and channel conditions also show that classifiers may learn collection-specific structure together with transmitter effects [6]–[8]. These findings make deployment robustness a central concern in practical RFFI [25]–[28].

B. OOB Evidence

Power-amplifier nonlinearity, oscillator behavior, and transmitter filtering may leave discriminative structure outside the nominal information band. WideScan combined IB and OOB distortion and reported improved discrimination among similar devices [5]. Other systems use spectral hardware features for device recognition [29], [30]. These studies motivate the use of OOB features but do not establish whether those features remain stable across different deployment conditions.

C. Cross-Receiver RFFI

Recent cross-receiver work seeks representations that transfer across acquisition domains. Receiver-adversarial methods aim to reduce receiver-specific information in learned representations [9]. Domain-adaptation methods use alignment, pseudo-labeling, source-free adaptation, or feature disentanglement [10], [13]–[15], [31]. Other work uses federated, contrastive, or Siamese objectives to exploit receiver and channel diversity [11], [12], [16], [32]. These approaches differ in supervision and deployment assumptions but share the goal of improving transfer. Our focus is diagnosis. We intervene on the spectral evidence used by a fixed classifier and test whether the resulting dependence persists on receivers withheld from development.

Domain adaptation provides a general framework for learning under distribution shift [33], while test-time entropy minimization (TENT) updates a deployed model using target data [34]. Our predictor remains frozen during diagnosis. We target a defined evidence pathway, aggregate by acquisition domain, and fix the confirmation set before evaluation.

D. Positioning and Methodological Gap

Table I summarizes the methodological differences between this work and representative prior studies. Three constraints motivate the proposed design. First, a change in a frozen classifier's output or accuracy under a targeted digital OOB perturbation provides evidence of dependence on that pathway, although transmitter, channel, and receiver effects remain entangled in the captured waveform. Second, the intervention target must match the representation because a full-spectrum feature may contain both IB and OOB bins. Third, development findings should be confirmed on receivers that did not influence checkpoint, operator, or threshold selection.

III. System Model and Problem Formulation

A. Cross-Receiver LoRa RFFI Model

Let $\mathcal{Y}$ denote the set of ten transmitter classes, and let $\mathcal{R}$ denote the set of twenty receiver domains. Their elements are indexed by y and r, respectively. The development and blind sets, $\mathcal{R}_{\mathrm{dev}}$ and $\mathcal{R}_{\mathrm{blind}}$, contain fourteen and six receivers, respectively. At time t, $s_y(t)$ denotes the ideal complex waveform, g_y denotes the transmitter response, and $u_y(t)$ denotes the resulting distorted waveform. Here, $h_{y,r}$ denotes the channel impulse response, q_r denotes the receiver radio-frequency front end, n_r denotes additive noise, and $\mathcal{A}_r$ denotes the analog-to-digital conversion (ADC) and sampling operator:

$$u_y(t) = g_y(s_y(t)), \tag{1}$$

$$z_{y,r}(t) = q_r(h_{y,r}(t) * u_y(t)) + n_r(t), \tag{2}$$

$$x_{y,r}[n] = \mathcal{A}_r\{z_{y,r}(t)\}. \tag{3}$$

Here, $*$ denotes convolution, $z_{y,r}(t)$ denotes the pre-ADC signal, n denotes the sample index, and $x_{y,r}[n]$ denotes the captured complex I/Q packet. Fig. 1 places the digital IB and OOB views after the ADC. We do not attempt to recover the individual analog operators from the captured signal.

B. IB and OOB Definitions

Let $L = 8192$ denote the number of complex samples in each packet, and let $F_s = 1$ MHz denote the sampling frequency. Each packet is normalized as

$$x[n] = \frac{x_{y,r}[n]}{\sqrt{L^{-1}\sum_{\ell=0}^{L-1} |x_{y,r}[\ell]|^2 + \epsilon_x}}, \qquad \epsilon_x = 10^{-8}, \tag{4}$$

where ℓ is a summation index and ϵ_x prevents division by zero. The L-point discrete Fourier transform (DFT), computed by

TABLE I

POSITIONING RELATIVE TO REPRESENTATIVE RFFI STUDIES. AN EVIDENCE INTERVENTION DENOTES A POST-TRAINING MODIFICATION OF A DEFINED INPUT PATHWAY. BLIND-RECEIVER CONFIRMATION REQUIRES RECEIVERS WITHHELD FROM CHECKPOINT, PREPROCESSING, OPERATOR, AND CRITERION SELECTION. MULTI-RECEIVER USE ALONE DOES NOT SATISFY THIS REQUIREMENT. A DASH MARKS A DIMENSION OUTSIDE THE CITED WORK'S PRINCIPAL CONTRIBUTION.

Work	OOB evidence	Channel shift	Receiver shift	Primary focus/strategy	Evidence intervention	Blind-receiver confirmation
ORACLE [2]	–	✓	–	augmentation	–	–
Exposing the Fingerprint [7]	–	✓	–	diagnosis	–	–
WideScan [5]	✓	limited	–	feature use	–	–
Shen *et al.* [9]	–	✓	✓	adversarial learning	–	multi-receiver
Ma *et al.* [16]	–	✓	✓	Siamese/contrastive	–	multi-receiver
This work	✓	–	✓	**dependence diagnosis**	✓	**six blind receivers**

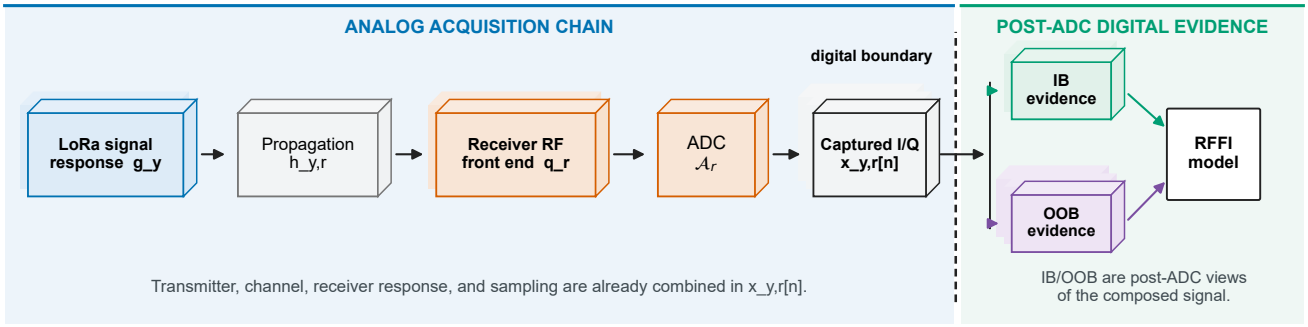


Fig. 1. Received-signal model and evidence boundary. The transmitter, propagation channel, receiver front end, and analog-to-digital converter (ADC) act before the captured I/Q packet is divided into digital IB and OOB evidence. A transmitter-related component may therefore remain useful even when its acquired representation changes with the receiver.

fast Fourier transform (FFT) and arranged with `fftshift`, is

$$X[k] = \mathrm{DFT}_L\{x[n]\}, \qquad k = 0, \ldots, L-1. \tag{5}$$

Here, $f_k = (k - L/2)F_s/L$ is the baseband frequency of bin k, with 122.07-Hz spacing. For nominal LoRa bandwidth $B = 125$ kHz,

$$\mathcal{K}_{\mathrm{IB}} = \{k : |f_k| \le B/2\}, \qquad \mathcal{K}_{\mathrm{OOB}} = \{k : |f_k| > B/2\}, \tag{6}$$

where $|\mathcal{K}|$ denotes the cardinality of set $\mathcal{K}$. The signal audit uses

$$R_{\mathrm{OOB}} = \sqrt{\frac{1}{|\mathcal{K}_{\mathrm{OOB}}|} \sum_{k \in \mathcal{K}_{\mathrm{OOB}}} |X[k]|^2}, \tag{7}$$

$$R_{\mathrm{IB}} = \sqrt{\frac{1}{|\mathcal{K}_{\mathrm{IB}}|} \sum_{k \in \mathcal{K}_{\mathrm{IB}}} |X[k]|^2}, \tag{8}$$

$$\rho_{\mathrm{OOB}} = 20 \log_{10} \frac{R_{\mathrm{OOB}} + \epsilon}{R_{\mathrm{IB}} + \epsilon}, \qquad \epsilon = 10^{-8}. \tag{9}$$

R_{OOB} and R_{IB} denote the root-mean-square (RMS) magnitudes over their respective bin sets, and ρ_{OOB} denotes their ratio in decibels (dB). The stabilizer $\epsilon = 10^{-8}$ prevents division by zero.

C. Receiver-Sensitive OOB Dependency

Let a index the representation family, and let θ denote its frozen classifier parameters. For non-targeted evidence $N^{(a)}(x)$ and the family-specific OOB target $E^{(a)}_{\mathrm{OOB}}(x)$, the prediction is

$$\widehat{y} = f^{(a)}_\theta \left(N^{(a)}(x), E^{(a)}_{\mathrm{OOB}}(x) \right), \qquad \hat{y} \in \mathcal{Y}. \tag{10}$$

Under condition c, operator T_c is applied only to the targeted OOB evidence:

$$\widehat{y}_c = f^{(a)}_\theta \left(N^{(a)}(x), T_c(E^{(a)}_{\mathrm{OOB}}(x)) \right). \tag{11}$$

For LateFusion, the intervention changes the complex-FFT OOB bins and explicit OOB-magnitude view. For CrossAttn, it changes the OOB keys and values. Labels, checkpoints, and non-targeted inputs remain unchanged.

We use *OOB utility* to denote the performance difference between an OOB-aware pathway and its raw-spectrum IB counterpart. *OOB sensitivity* denotes the response of the frozen model to T_c, and *receiver transfer* denotes clean performance on an unseen physical receiver. These quantities describe different properties and need not agree.

D. Receiver-Level Diagnostic Quantities

Let i index packets, let $m = 1, \ldots, M$ index training seeds, and let c index experimental conditions. Let N_r denote the

TABLE II
MATHEMATICAL NOTATION USED IN THIS WORK.

Symbol	Meaning
$\mathcal{Y}, \mathcal{R}, \mathcal{R}_{\mathrm{dev}}, \mathcal{R}_{\mathrm{blind}}$	class/receiver sets and development/blind subsets
y, r, t, n	class, receiver, time, and sample indices
$s_y, g_y, u_y, h_{y,r}, *, q_r, n_r$	waveforms, channel/convolution, front end, and noise
$z_{y,r}, \mathcal{A}_r, x_{y,r}, x$	pre-ADC signal, sampler, packet, and shorthand
$L, F_s, B, k, f_k, X[k]$	length/rate/bandwidth and bin/frequency/spectrum
$\mathcal{K}_{\mathrm{IB}}, \mathcal{K}_{\mathrm{OOB}}, \lvert\mathcal{K}\rvert$	IB/OOB bin sets and set cardinality
$R_{\mathrm{IB}}, R_{\mathrm{OOB}}, \rho_{\mathrm{OOB}}, \epsilon, \epsilon_x$	IB/OOB RMS, relative magnitude, and stabilizers
$a, \theta, f_\theta^{(a)}$	family, parameters, and classifier
$N^{(a)}, E_{\mathrm{OOB}}^{(a)}$	non-targeted and targeted evidence
$c, T_c, \widehat{y}, \widehat{y}_c, \alpha, T_\alpha$	condition/operator/predictions and scale/operator
$E_i, E_j, \bar{E}_{\mathrm{dev}}, \mathbf{o}_i$	target/donor/neutral evidence and OOB vector
$T_{\mathrm{shuf}}, T_{\mathrm{neu}}, T_L, T_R, \mathcal{T}$	shuffle/neutral/side operators and operator set
$\mathcal{D}_r, \mathcal{D}_r^{(256)}$	evaluation set and 256-packet template subset
$i, m, M, N_r, \mathbf{1}(\cdot)$	packet/seed indices, counts, and indicator
$A_{r,m,c}, \bar{A}_{r,c}, \Delta_{r,c}$	accuracy/seed mean/clean-minus-condition drop (pp)
$\Delta_{r,\mathrm{scale}}, \Delta_{r,\mathrm{disrupt}}$	worst-scale and disruption effects
$\Delta_{r,\mathrm{L-R}}$	left-minus-right spectral effect

number of evaluated packets for receiver r, and let $\mathbf{1}(\cdot)$ denote the indicator function. We define

$$A_{r,m,c} = \frac{1}{N_r} \sum_{i=1}^{N_r} \mathbf{1}(\widehat{y}_{i,r,m,c} = y_i), \tag{12}$$

$$\bar{A}_{r,c} = \frac{1}{M} \sum_{m=1}^{M} A_{r,m,c}, \tag{13}$$

$$\Delta_{r,c} = 100 \left(\bar{A}_{r,\mathrm{clean}} - \bar{A}_{r,c} \right). \tag{14}$$

Here, y_i and $\widehat{y}_{i,r,m,c}$ are the true and predicted labels. The effect $\Delta_{r,c}$ is in percentage points (pp), and $\bar{A}_{r,c}$ is the receiver-level summary for condition c.

In this paper, *receiver-sensitive* denotes the coexistence of receiver-associated information and effects that vary across receivers. This operational definition does not imply hardware isolation or a receiver-by-condition causal effect.

IV. Proposed Receiver-Sensitive OOB Diagnostic Framework

A. Method Overview and Scope

Table II summarizes the notation used throughout the paper.

Fig. 2 summarizes the signal audit, frozen interventions, controls, and blind confirmation. During each intervention, model parameters and non-targeted inputs remain fixed, so the measured response reflects the prescribed change to the targeted OOB pathway. All design choices are fixed before blind evaluation. Comparison of the two representation families tests whether the observed dependence is specific to one computational interface. A drop under (11) provides evidence of dependence on the targeted digital pathway, but it does not identify the physical source of that evidence. TrueIB instead modifies the full feature-generation pathway.

B. Diagnostic Representation Families

Fig. 3 shows the two diagnostic families.

1) LateFusion-OOB: LateFusion-OOB has four one-dimensional encoders for I/Q (two channels), complex FFT (two), amplitude/phase (two), and OOB magnitude (one). Each encoder consists of three convolutional stages with 32, 64, and 128 channels. Batch normalization (BN), sigmoid linear unit (SiLU) activation, and pooling are applied within these stages, and each encoder outputs a 128-dimensional vector. A multilayer perceptron (MLP) maps the concatenated 512-dimensional representation through $512 \to 512 \to 128$ to ten classes. Interventions modify the complex-FFT OOB bins and explicit OOB-magnitude view while leaving I/Q and amplitude/phase unchanged. The probe has 539,178 parameters.

2) CrossAttn-OOB: CrossAttn-OOB conditions a main representation on OOB evidence. It maps 8192 I/Q samples to 32 main tokens. Its five-channel stem receives I, Q, FFT magnitude, instantaneous amplitude, and phase. Convolutions with kernel sizes 7 and 5 are followed by a convolution whose kernel size and stride are both 256, yielding $Z_m \in \mathbb{R}^{32\times64}$. The 8192-point masked spectrum is normalized by the IB RMS and divided into 32 contiguous 256-bin patches. The patches are embedded as $Z_o \in \mathbb{R}^{32\times64}$. After layer normalization (LN), multi-head attention (MHA) uses Z_m as the query and Z_o as the key and value:

$$Z_a = \mathrm{MHA}(Q = \mathrm{LN}(Z_m), K = \mathrm{LN}(Z_o), V = \mathrm{LN}(Z_o)). \tag{15}$$

The attended representation enters an elementwise gate,

$$G = \sigma(W_g \,\mathrm{Concat}(Z_m, Z_a) + b_g), \tag{16}$$

$$\widetilde{Z}_m = \mathrm{LN}(Z_m + G \odot W_o Z_a), \tag{17}$$

where σ is the logistic sigmoid, Concat concatenates along the feature dimension, W_g and b_g are the gate parameters, W_o projects the attended output, and $\odot$ is the Hadamard product. The fused representation then passes through two mixers, a final LN layer, mean pooling, and a ten-class classification head. Chirp embedding is disabled. The model has 625,896 parameters. Interventions modify Z_o and the corresponding K and V while leaving main-token construction and the checkpoint fixed.

3) Relation to HSTU: The downstream mixers adapt the Hierarchical Sequential Transduction Unit (HSTU) [35]. For input Z_h, one projection forms query, key, value, and gate branches q_h, k_h, v_h, u_h. With query/key dimension d_h and learned relative-position bias B_{rel},

$$\Pi_h = \sigma\left(q_h k_h^{\mathsf{T}} / \sqrt{d_h} + B_{\mathrm{rel}} \right), \tag{18}$$

$$H_h = \mathrm{LN}(\Pi_h v_h) \odot \mathrm{SiLU}(u_h), \tag{19}$$

$$Z_h' = \mathrm{LN}(Z_h + W_h H_h). \tag{20}$$

Here, Π_h is the sigmoid mixing matrix, H_h is the gated output, W_h is the output projection, and Z_h' is the updated sequence. The noncausal mixer uses sigmoid rather than softmax. In this work, HSTU is adapted to fixed-length RF patches and ten-device classification. Only its gated token-mixing mechanism is retained.

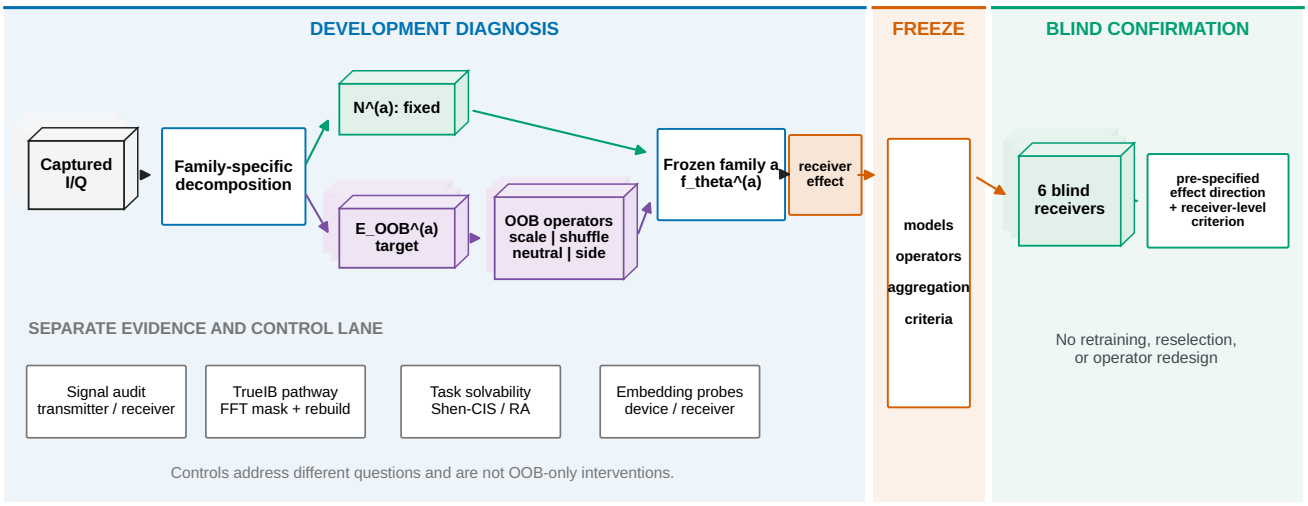


Fig. 2. Receiver-sensitive OOB dependency diagnostic framework. $N^{(a)}$ and $E^{(a)}_{\mathrm{OOB}}$ denote non-targeted and targeted evidence. TrueIB is the strict raw-spectrum in-band control. CIS and RA denote the channel-independent spectrogram and receiver-adversarial training, respectively. All development choices are fixed before blind confirmation.

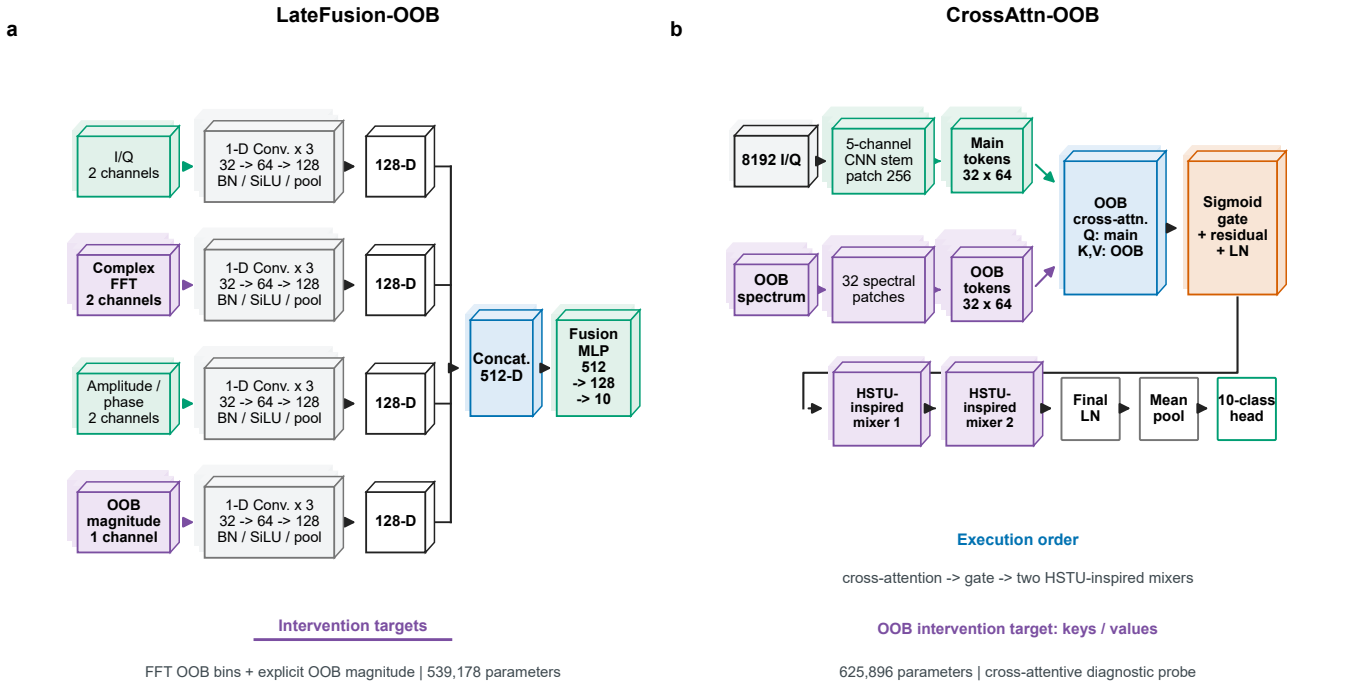


Fig. 3. Architectures of the diagnostic probes. (a) LateFusion-OOB encodes I/Q, complex FFT, amplitude/phase, and OOB magnitude using one-dimensional convolutional encoders. (b) CrossAttn-OOB uses a five-channel CNN stem, OOB cross-attention, a sigmoid gate, and two HSTU-inspired mixers. BN, SiLU, MLP, LN, and HSTU denote batch normalization, sigmoid linear unit, multilayer perceptron, layer normalization, and Hierarchical Sequential Transduction Unit.

C. TrueIB Pathway Controls

TrueIB is the strict raw-spectrum in-band control. The mask is applied to the spectrum of normalized raw I/Q samples rather than to downstream features:

$$X_{\mathrm{TrueIB}}[k] = X[k]\mathbf{1}(|f_k| \le 62.5\ \mathrm{kHz}), \tag{21}$$

$$x_{\mathrm{TrueIB}}[n] = \mathrm{IFFT}\{X_{\mathrm{TrueIB}}[k]\}. \tag{22}$$

The inverse fast Fourier transform (IFFT) reconstructs $x_{\mathrm{TrueIB}}[n]$, from which all downstream views are regenerated.

LateFusion-TrueIB (539,178 parameters) is development-only, whereas CrossAttn-TrueIB (580,008 parameters) is also evaluated on blind receivers. Shen-CIS and Shen-RA test whether identification remains solvable on held-out receivers when a receiver-robust representation is used (Section V). They are task-solvability controls, not OOB interventions or capacity-matched competitors.

D. Frozen OOB Interventions

All operators retain labels, checkpoints, and non-targeted evidence. For family-specific OOB evidence E, the scaling operator is

$$T_\alpha(E) = \alpha E, \quad \alpha \in \left\{\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2\right\}, \tag{23}$$

where $\alpha = 1$ corresponds to the unmodified condition. Same-receiver cross-device shuffle uses a deterministic donor j for packet i:

$$T_{\text{shuf}}(E_i) = E_j, \qquad r_j = r_i, \quad y_j \neq y_i. \tag{24}$$

The donor shares the receiver but has a different class label. Neutral replacement uses

$$T_{\text{neu}}(E_i) = \bar{E}_{\text{dev}}. \tag{25}$$

For each $r \in \mathcal{R}_{\text{dev}}$, $\mathcal{D}_r^{(256)}$ contains 256 template packets. Complex RMS normalization with a 10^{-8} stabilizer and shifted-FFT OOB extraction give $\mathbf{o}_i \in \mathbb{R}_+^{7167}$. Here, $7167 = 8192 - 1025$ is the number of OOB bins for $L = 8192$, $F_s = 1$ MHz, and $|f| > 62.5$ kHz, and $\mathbb{R}_+$ denotes the nonnegative reals. The vector contains magnitudes only. The frozen template is

$$\bar{E}_{\text{dev}} = \frac{1}{|\mathcal{R}_{\text{dev}}|} \sum_{r \in \mathcal{R}_{\text{dev}}} \frac{1}{256} \sum_{i \in \mathcal{D}_r^{(256)}} \mathbf{o}_i. \tag{26}$$

Equation (26) averages within each receiver and weights all fourteen receivers equally. The resulting template is real and nonnegative, with zero phase, and is fixed before blind evaluation. T_L and T_R attenuate bins below -62.5 kHz and above $+62.5$ kHz by 0.5 without renormalization. Left and right attenuation are treated separately because the complex baseband spectrum need not be symmetric.

Scaling measures sensitivity to OOB strength. Same-receiver shuffle changes device-associated OOB content while retaining the receiver domain. Neutral replacement removes packet-specific OOB structure without importing another device sample. Left/right attenuation measures the frequency structure of the learned dependence.

E. Random OOB-Scale Augmentation

Random OOB-Scale Augmentation (RSA) targets OOB-scale sensitivity. At each update, $\log \alpha$ is sampled from $\mathcal{U}(\log 0.5, \log 2)$, where $\mathcal{U}(a, b)$ denotes the uniform distribution on $[a, b]$. The cross-entropy (CE) loss is

$$\mathcal{L}_{\text{RSA}} = \text{CE}\Big(f_\theta^{(a)}(N^{(a)}(x), T_\alpha(E_{\text{OOB}}^{(a)}(x))), y\Big). \tag{27}$$

No paired clean/scaled loss, consistency term, or IB scaling is used. Matched clean continuation (CT) starts from the same checkpoint and receives the same number of updates.

Algorithm 1 Locked receiver-level OOB diagnosis

Require: Frozen checkpoints $f_{\theta_m}^{(a)}$, intervention set $\mathcal{T}$, receiver packet sets $\{\mathcal{D}_r\}$, and fixed confirmation criteria
1: **for** each development receiver r and seed m **do**
2: Evaluate clean and intervention accuracies
3: **end for**
4: Aggregate seeds within each receiver
5: Freeze checkpoints, preprocessing, $\mathcal{T}$, criteria, and blind receiver list
6: **for** each blind receiver r and frozen checkpoint **do**
7: Evaluate 2,000 packets under all fixed conditions
8: **end for**
9: Compute $\bar{A}_{r,c}$ and $\Delta_{r,c}$ for each receiver
10: Compute receiver medians, ranges, and direction counts
11: Assess the receiver-level effects against the fixed criteria
Ensure: Receiver-level development diagnosis and blind confirmation

F. Frozen Blind-Receiver Procedure and Criteria

Algorithm 1 summarizes the development-to-blind evaluation procedure. Before blind evaluation, we fixed 25 checkpoints (five families with five seeds each), intervention constants, receiver partition, aggregation, and confirmation criteria. The primary criterion requires positive disruption and worst-scale effects on at least four of six receivers and a receiver median of at least 5 pp for each effect. The extended criterion additionally requires a positive left-minus-right effect on at least four receivers and a positive median. These finite-set engineering criteria do not provide frequentist error control. Accuracy is averaged over seeds within each receiver, and seeds are not treated as receiver replicates.

Once blind evaluation begins, no retraining, checkpoint reselection, operator redesign, or receiver exclusion is allowed. All six receiver results are reported. For receiver r, $\mathcal{D}_r$ denotes the evaluation set, θ_m denotes the seed-m checkpoint, and $\mathcal{T} = \{T_\alpha, T_{\text{shuf}}, T_{\text{neu}}, T_L, T_R\}$ denotes the fixed intervention set.

V. Experimental Setup

A. Dataset and Receiver Partition

We use the public cross-receiver LoRa corpus collected with twenty software-defined radio (SDR) receivers [9]. It contains ten transmitter classes and twenty receiver domains. Fourteen receivers form $\mathcal{R}_{\text{dev}}$, with 8,000 training packets per receiver. Six files form $\mathcal{R}_{\text{blind}}$: b200_2, b200_mini_2, b210_2, n210_2, n210_3, and pluto_2. Each blind receiver contributes 2,000 balanced packets, with 200 packets per class. We evaluate closed-set identification. Unknown-device rejection is outside the scope of this study.

The signal audit covers 24 Hierarchical Data Format version 5 (HDF5) files, 144,000 packets, and 240 receiver–device–file cells. Some receiver instances contribute repeated or day-specific files, so the 24 files are not 24 independent receiver domains. The blind inventory and archive hash were recorded before any signal values from the blind files were examined. Blind receivers were excluded from training, validation, checkpoint selection, preprocessing design, neutral-template construction, operator selection, and threshold selection.

B. Signal Processing

Each packet contains 8192 complex samples at $F_s = 1$ MHz and is normalized according to (4), including the 10^{-8}

TABLE III
EVALUATED MODELS AND EXPERIMENTAL ROLES. PARAMETER COUNTS REFER TO THE IMPLEMENTATIONS USED HERE.

Model	Principal input	Spectral pathway	Parameters	Role
LateFusion-OOB	four RF views	targeted FFT/OOB views	539,178	convolutional dependency probe
CrossAttn-OOB	main and OOB tokens	OOB keys/values	625,896	token dependency probe
LateFusion-TrueIB	regenerated IB views	raw-spectrum IB mask	539,178	development-only pathway control
CrossAttn-TrueIB	regenerated IB views	raw-spectrum IB mask	580,008	development + blind pathway control
Shen-CIS	52×126 CIS	–	13,510,424	task-solvability control
Shen-RA	CIS + receiver labels	gradient reversal	13,510,424	receiver-adversarial control

TABLE IV
TRAINING SETTINGS AND FIXED CHECKPOINTS FOR BLIND CONFIRMATION. LR, WD, AND SGD DENOTE LEARNING RATE, WEIGHT DECAY, AND STOCHASTIC GRADIENT DESCENT. BLIND RECEIVERS WERE NOT USED FOR EPOCH SELECTION.

Model	Optimizer	LR	WD/momentum	Batch	Development use	Blind checkpoint rule
LateFusion-OOB	AdamW	10^{-3}	WD 5×10^{-4}	64	6 folds, 2 seeds	fixed epoch 5
CrossAttn-OOB	AdamW	10^{-3}	WD 5×10^{-4}	64	6 folds, 2 seeds	fixed epoch 4
CrossAttn-TrueIB	AdamW	10^{-3}	WD 5×10^{-4}	64	6 folds, 2 seeds	fixed epoch 5
Shen-CIS	SGD	10^{-3}	momentum 0.9	64	6 folds, 2 seeds	fixed epoch 74
Shen-RA	SGD	10^{-3}	momentum 0.9	64	6 folds, 2 seeds	fixed epoch 71

stabilizer. An 8192-point FFT followed by `fftshift` places the IB boundary at ± 62.5 kHz. The remaining bins form the OOB region in (6). All model views use the normalized packet. Only TrueIB masks the raw spectrum and reconstructs downstream views.

C. Experimental Roles

Table III distinguishes dependency probes, TrueIB controls, and task-solvability controls. Shen-CIS uses a 52×126 channel-independent spectrogram (CIS), and Shen-RA adds a gradient-reversal receiver head trained with development labels [9], [36]. Both are PyTorch adaptations of the published TensorFlow implementations. They are used as task-solvability controls rather than as capacity-matched competitors.

D. Training and Checkpoint Selection

Development diagnosis uses six folds, each with one receiver held out, and two seeds per fold. Confirmatory models are trained on all fourteen development receivers with five seeds, without blind validation or early stopping. Diagnostic and TrueIB models use AdamW. The task-solvability controls use stochastic gradient descent (SGD). Table IV lists the fixed settings. CT and RSA start from the same checkpoint and receive the same number of updates. Only the training view differs.

E. Metrics and Receiver-Level Statistics

Accuracy, its average over seeds, and the clean-minus-condition effect are defined in (14). The non-unit scale set $\mathcal{A}' = \{1/2, 1/\sqrt{2}, \sqrt{2}, 2\}$ corresponds to approximately $-6, -3, +3, +6$ dB in magnitude. The worst-scale effect is

$$\Delta_{r,\mathrm{scale}} = 100 \left[\bar{A}_{r,\mathrm{clean}} - \min_{\alpha \in \mathcal{A}'} \bar{A}_{r,\alpha} \right]. \tag{28}$$

The disruption effect is

$$\Delta_{r,\mathrm{disrupt}} = 100 \left[\bar{A}_{r,\mathrm{clean}} - \frac{\bar{A}_{r,\mathrm{shuf}} + \bar{A}_{r,\mathrm{neu}}}{2} \right], \tag{29}$$

and spectral asymmetry is $\Delta_{r,\mathrm{L-R}} = \Delta_{r,L} - \Delta_{r,R}$. Macro-F1 is the unweighted mean of classwise F1 scores and is averaged over seeds within each receiver.

For the development experiments, packet accuracy is first averaged over two seeds within each held-out receiver and then summarized across the six folds. Blind results use five seeds for each receiver. We report receiver medians, means, ranges, and direction counts without packet-pooled confidence intervals. Signal-audit SDs describe group-mean spread, not a variance decomposition.

F. Frozen Embedding Probes

Frozen embeddings are averaged over 256 packets within each receiver–device cell. Standardization uses probe-training cells only. A linear softmax classifier is optimized for 300 steps using Adam, with a learning rate of 0.05 and weight decay of 10^{-3}. In each development fold, the held-out receiver is excluded from embedding extraction, leaving 13 receiver groups. Device probes are trained on a 75% subset of receiver groups selected using a fixed random seed and evaluated on the remainder. Receiver probes are trained on seven devices and evaluated on three held-out devices while classifying all 13 receiver groups. Results are averaged across six folds and two seeds. Chance accuracy is 10.0% for devices and $1/13 = 7.7\%$ for receivers. Probe accuracy measures decodability. It does not establish feature purity or causal attribution.

G. Blind Evaluation Workload and Implementation

Five model families, excluding development-only LateFusion-TrueIB, are evaluated across five seeds and six blind receivers, giving 150 clean runs. The two OOB-aware families are additionally evaluated under eight interventions: four non-unit scales, shuffle, neutral replacement, left attenuation, and right attenuation. This yields 480 intervention runs and 630 runs in total, with 2,000 packets in each run. Experiments use Python 3.12.12, PyTorch 2.10.0, CUDA 12.9, NumPy 2.4.4, and h5py 3.16.0

TABLE V
FROZEN OOB-EMBEDDING PROBE ACCURACY (%).

Probe	LateFusion	CrossAttn	Chance
Device (receivers held out)	94.4	97.9	10.0
Receiver (devices held out)	16.2	14.7	7.7

on an NVIDIA A100-SXM4-80GB graphics processing unit (GPU).

VI. EXPERIMENTAL RESULTS AND ANALYSIS

Results are organized around six research questions (RQs).

A. RQ1: Does the Captured OOB Region Contain Transmitter and Receiver Structure?

Fig. 4 shows group means of ρ_{OOB} for ten devices, six receiver types, fourteen receiver instances, and five day labels. The descriptive standard deviation (SD) across device means is 0.393 dB. The corresponding SDs for receiver type, receiver instance, and day are 0.111, 0.083, and 0.015 dB. Because the groups differ in size and hierarchy, these values describe group-mean spread rather than variance components. The OOB measurement therefore varies across devices and also contains acquisition-associated structure.

B. RQ2: Do Frozen Classifiers Depend on the Targeted OOB Pathway?

Fig. 5 summarizes the receiver-level scale and shuffle effects on the development folds. Across the six development folds, each with one receiver held out, the median worst-scale drops are 47.28 pp for LateFusion-OOB and 56.60 pp for CrossAttn-OOB. Same-receiver cross-device shuffle produces 47.39 and 60.90 pp, and neutral replacement produces 43.94 and 56.77 pp. The comparable shuffle and neutral responses indicate that the finding is not specific to one replacement scheme. Because the shuffle donor comes from the same receiver but a different transmitter, this pattern is consistent with device-associated OOB content contributing to the classifier's decision. The measured drop may also include cross-view incompatibility because non-targeted views retain the original packet.

Frozen-embedding probes provide complementary evidence (Table V). Device labels are highly decodable with receivers held out, while receiver-group classification accuracy remains above the 7.7% chance level with devices held out. These results indicate that the OOB embeddings contain transmitter information while retaining receiver-associated information.

C. RQ3: Is OOB Dependence Representation-Independent?

The controls in Fig. 6(a) show different patterns across families. Mean accuracy on the held-out development receivers is 56.64% for LateFusion-OOB and 11.97% for LateFusion-TrueIB. The ordering reverses for CrossAttn: OOB reaches 60.90%, whereas TrueIB reaches 70.91%. OOB is therefore neither uniformly beneficial nor uniformly harmful across representations.

Shen-CIS and Shen-RA reach 94.10% and 93.85% accuracy, respectively. Their paired receiver-mean difference (RA minus CIS) is -0.25 pp, with two positive differences, three negative differences, and one zero difference. These 13.51-million-parameter controls show that the identification task remains solvable when receivers are held out, although their capacity differs from that of the diagnostic probes.

D. RQ4: Does Controlled Scale Invariance Transfer to Unseen Receivers?

On the development folds, RSA reduces the mean accuracy drop caused by halving the OOB amplitude ($\alpha = 0.5$) from 53.96 pp under CT to 2.92 pp, as shown in Fig. 6(b). This confirms that RSA suppresses the intended scale sensitivity. Clean accuracy on held-out receivers does not improve. LateFusion changes from 64.85% under CT to 61.69% under RSA, and CrossAttn changes from 63.66% to 60.02%. For CrossAttn, the median paired accuracy change across receivers is -3.88 pp, as shown in Fig. 6(c). Robustness to this OOB-scale nuisance therefore does not imply invariance to receiver replacement.

E. RQ5: Is the Learned Dependence Frequency-Structured?

On the development folds, the mean left-OOB drops are 44.41 pp for LateFusion-OOB and 51.12 pp for CrossAttn-OOB, whereas right-OOB drops are 17.04 and 1.12 pp. These are receiver means, not medians. The consistently larger left-side effect motivated the fixed blind criterion. Because a complex baseband spectrum need not be symmetric, the intervention reveals frequency-dependent structure in the learned dependence without identifying its analog source.

F. RQ6: Do the Development Findings Replicate on Blind Receivers?

We evaluated the six blind receivers only after fixing the checkpoints, operators, aggregation rules, and confirmation criteria. LateFusion-OOB has median disruption and worst-scale drops of 49.05 and 51.83 pp. The corresponding CrossAttn-OOB medians are 71.78 and 70.66 pp. Both effects are positive on all six receivers for both representations, satisfying the primary criterion.

Left-minus-right effects are also positive on all six receivers, with medians of 29.89 pp for LateFusion-OOB and 70.56 pp for CrossAttn-OOB. Both families therefore meet the extended criterion. The blind evaluation confirms the direction and receiver-level consistency of the dependency pattern. These results describe the defined digital pathway rather than its physical source.

Table VI shows that Shen-CIS and Shen-RA achieve 96.41–100.00% accuracy on the blind receivers, whereas the diagnostic probes and CrossAttn-TrueIB show representation-dependent transfer. Thus, large OOB intervention effects coexist with the high accuracy achieved by the receiver-robust controls on the same identification task. Exact receiver-level values for Fig. 7 are provided with the manuscript source.

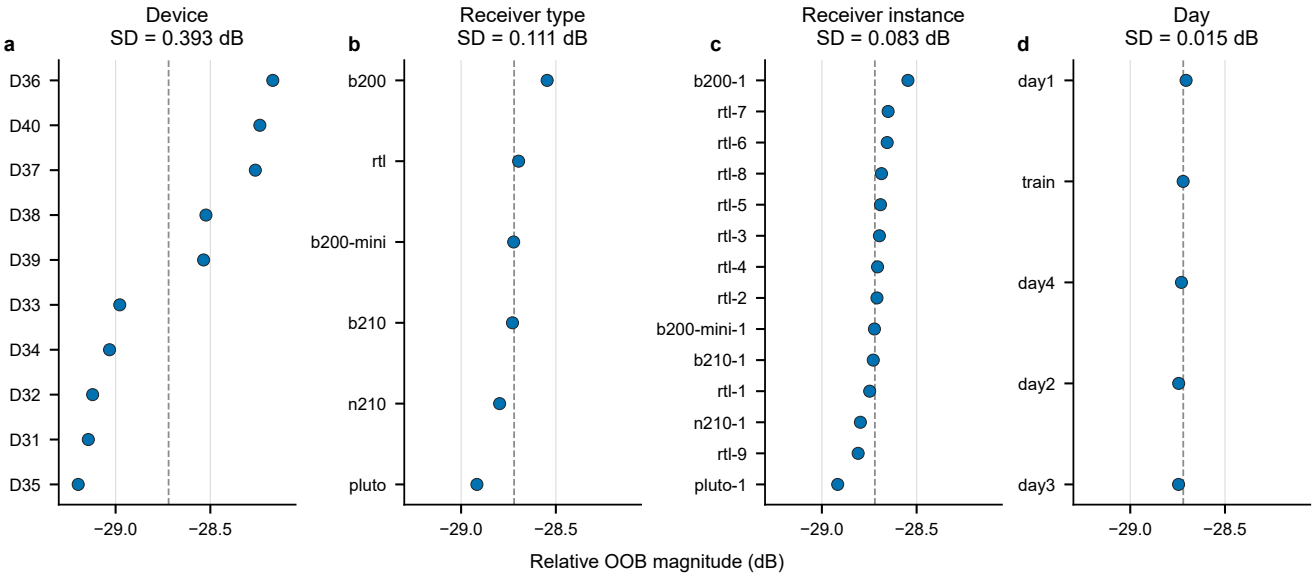


Fig. 4. Descriptive signal audit of relative OOB magnitude. Each marker is a group mean. D31–D40 denote the ten transmitter identities, and dashed lines show the overall mean. SD annotations describe group-mean spread rather than a variance decomposition.

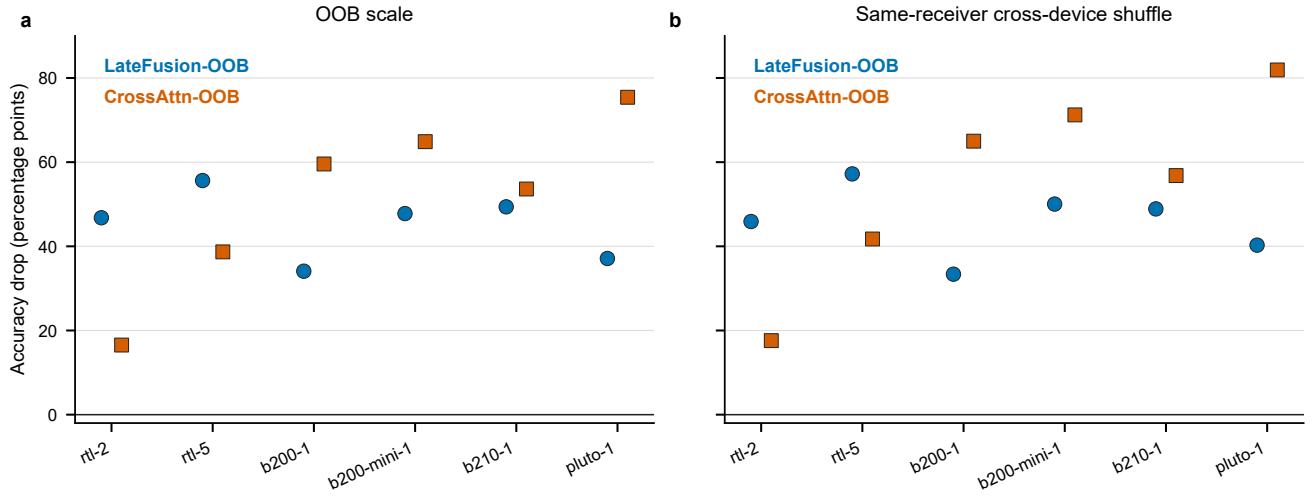


Fig. 5. Development receiver-level intervention effects. Each marker is the two-seed mean for one development fold with one receiver held out. Panels show clean-minus-condition accuracy drops for the worst non-unit OOB scale and same-receiver cross-device shuffle.

TABLE VI
BLIND CLEAN PERFORMANCE ACROSS SIX RECEIVERS (%). STATISTICS ARE COMPUTED ACROSS RECEIVER-LEVEL FIVE-SEED MEANS. SHEN-CIS AND SHEN-RA ARE TASK-SOLVABILITY CONTROLS WITH DIFFERENT INPUT PATHWAYS AND CAPACITY.

Metric	Statistic	Shen-CIS	Shen-RA	LateFusion-OOB	CrossAttn-OOB	CrossAttn-TrueIB
Accuracy	Mean	99.24	99.14	61.60	82.32	63.49
	Median	99.63	99.54	61.85	82.79	62.91
	Range	96.76–99.99	96.41–100.00	55.68–67.95	72.15–88.89	58.42–71.61
Macro-F1	Mean	99.23	99.12	57.13	80.78	60.64
	Median	99.63	99.54	57.27	81.61	59.68
	Range	96.69–99.99	96.27–100.00	50.27–64.06	69.62–88.22	55.43–69.35

G. Evidence Synthesis

Signal and embedding analyses show that transmitter identity is highly decodable while receiver-associated information remains present in the OOB representation. Shuffle and neutral replacement reduce accuracy, supporting dependence on packet-specific OOB inputs. Because non-targeted views retain

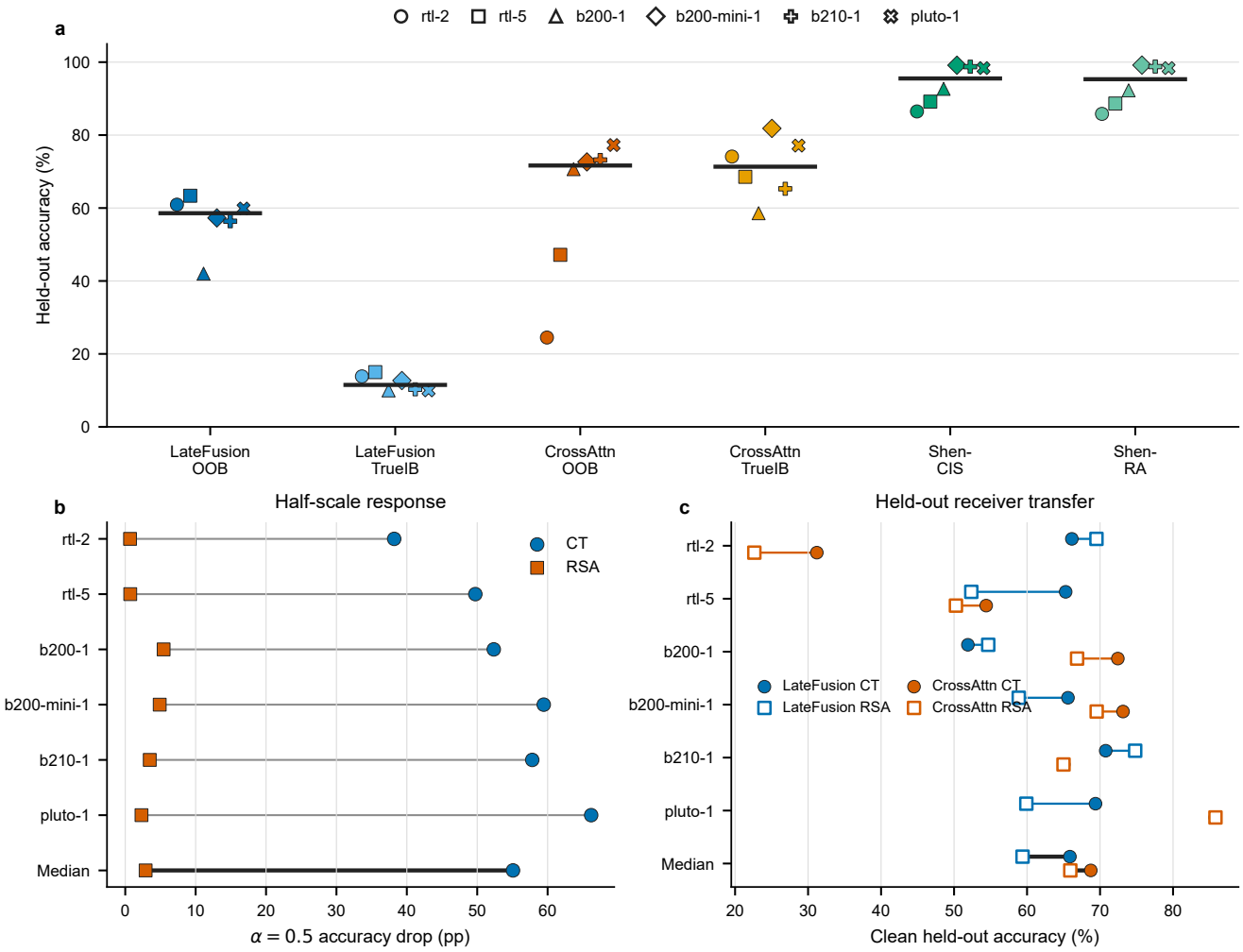


Fig. 6. Representation boundary and Random OOB-Scale Augmentation (RSA). (a) Development pathway and task-solvability controls. Markers are two-seed receiver means, and black segments are receiver medians. (b) $\alpha = 0.5$ accuracy drop under clean continuation (CT) and RSA. Each receiver value is averaged over two diagnostic backbones and two seeds. (c) Clean receiver-held-out accuracy under CT and RSA. Connected points are paired receiver units. The last row is the receiver median.

the original packet, the measured effects may also include cross-view incompatibility and are therefore interpreted as pathway dependence. Scale and spectral-side interventions show that accuracy depends on both OOB amplitude and the spectral region being modified.

TrueIB produces opposite utility patterns across model families, showing that OOB utility depends on the representation. CIS-based models solve the same task at high accuracy under receiver holdout, so split difficulty alone cannot explain the diagnostic results. RSA substantially reduces sensitivity to the tested scale perturbation, although clean transfer declines relative to matched continuation. All three predefined directional patterns appear on all six blind receivers after the protocol is fixed. These results indicate that OOB pathway dependence persists across receiver domains and coexists with receiver-associated information in the learned representation. They do not identify the physical source of that dependence or establish the same behavior for other representations.

VII. Discussion

A. Interpretation of OOB Dependence

The deployment issue is whether a representation uses OOB information consistently across receivers. Device probes and TrueIB controls do not support removing OOB by default, whereas the blind interventions show that OOB utility does not establish receiver invariance. These digital results can guide further RF measurements, but they do not identify a mixer, filter, oscillator, or receiver front end as the unique physical cause.

B. Controlled Robustness and Nuisance Specification

RSA reduces the mean response to the tested scale perturbation, yet matched accuracy on held-out receivers declines. Robustness to a specified transformation therefore differs from robustness to broader domain change. Robustness studies should specify the tested perturbation and whether the model is frozen or retrained. Claims of receiver generalization should be supported by evaluation on unseen physical receivers.

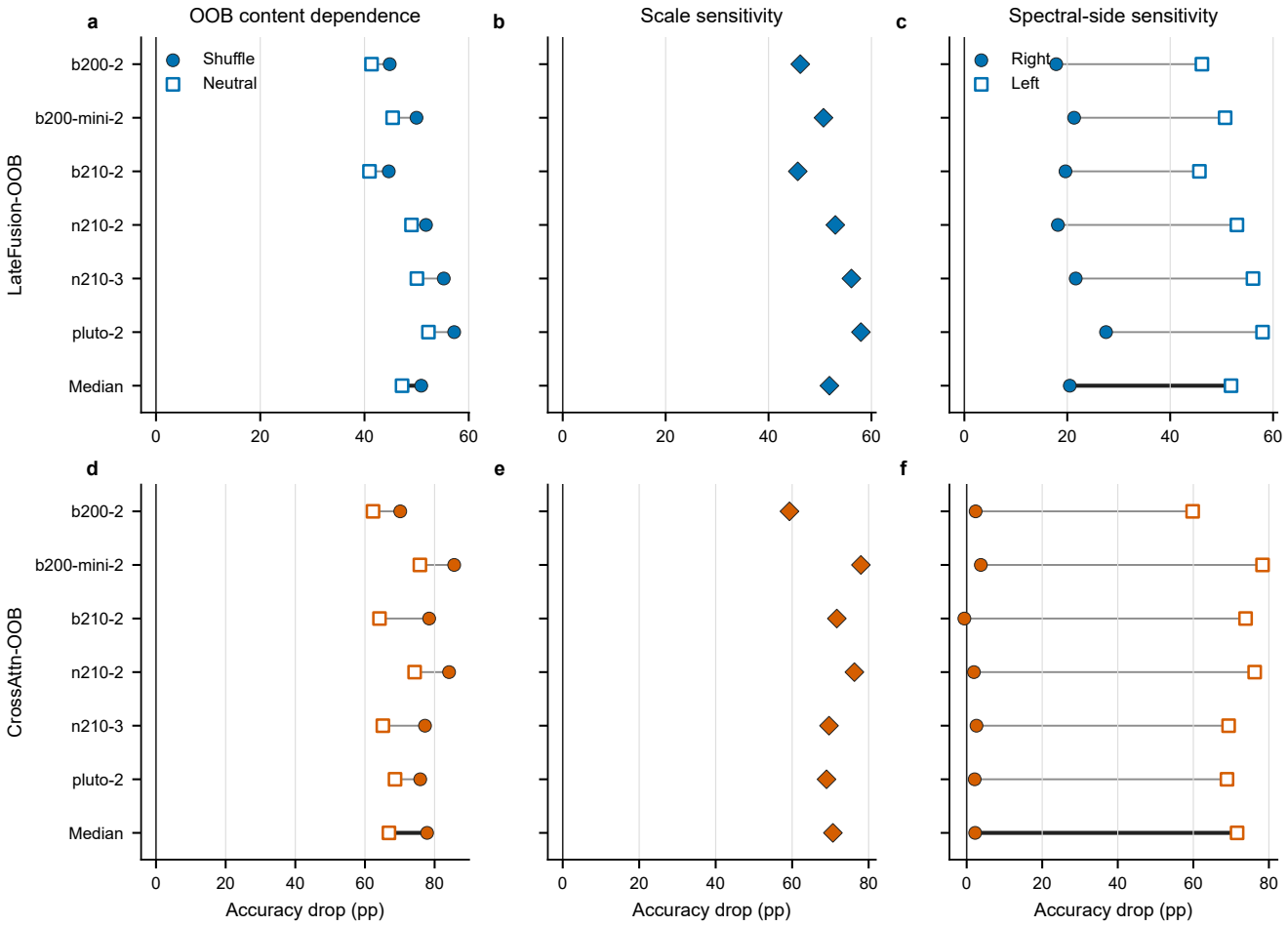


Fig. 7. Blind confirmation for LateFusion-OOB (top) and CrossAttn-OOB (bottom). Columns show OOB content dependence, worst-scale sensitivity, and left/right sensitivity. Receiver rows are five-seed means, and the final bold segment is the receiver median. Effects are clean-minus-condition accuracy drops in pp.

C. Implications for RFFI Evaluation

An OOB-aware evaluation should identify where OOB enters the representation, perturb that pathway without changing unrelated inputs, aggregate by receiver, and treat raw-spectrum TrueIB as a pathway control. Directional criteria should be fixed before results from the blind receivers are examined. This procedure complements physical calibration and transfer-learning studies by identifying dependencies that may guide subsequent experiments.

D. Limitations

This study is limited to one public corpus, closed-set identification of ten transmitters, six blind receivers, and the specified digital interventions. The released HDF5 files lack a complete capture hierarchy, so the signal audit cannot separate device, channel, day, and receiver components. The blind set covers only a limited range of gateways, bandwidths, propagation environments, and LoRa configurations, and the confirmation thresholds do not provide statistical guarantees for a broader receiver population. The digital interventions do not identify the analog source of the observed effects or eliminate all physical OOB influences. Responses may also include incompatibility between altered and non-targeted views. RSA addresses only random OOB scaling. Finally, closed-set identification does not establish open-set authentication. False acceptance rate, false rejection rate, equal error rate, and unknown-device rejection are outside the scope of this study.

VIII. Conclusion

OOB measurements can improve LoRa transmitter discrimination, but that utility does not imply stability under receiver replacement. We investigated this distinction by applying representation-specific interventions to frozen models, comparing pathway and task-solvability controls, and confirming the findings on blind receivers under a fixed protocol. Both diagnostic models relied on transmitter-discriminative OOB information, with intervention effects that varied across receivers and spectral regions. The predefined directional effects were observed on all six blind receivers. Random OOB-scale augmentation substantially reduced sensitivity to the tested scale perturbation but did not improve accuracy on held-out receivers relative to matched clean continuation. Robustness to an engineered OOB nuisance should therefore not be

interpreted as receiver invariance without evaluation on unseen physical receivers.

Data and Code Availability

The received-signal corpus is publicly available through the sources cited in Section V. No new primary dataset was created. Figure-source tables, generation scripts, code, experiment configurations, and archived results are available with the manuscript source and at https://github.com/simizhanghao/keyan.

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